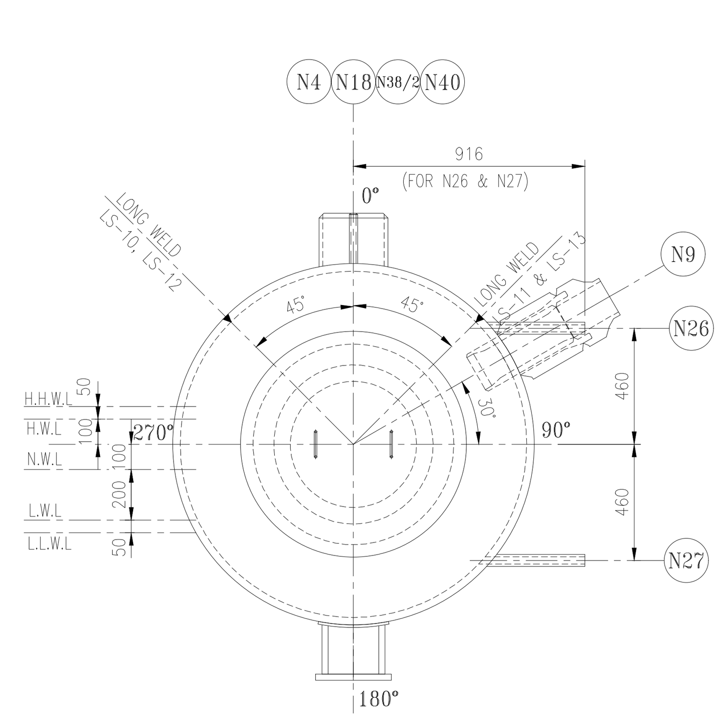
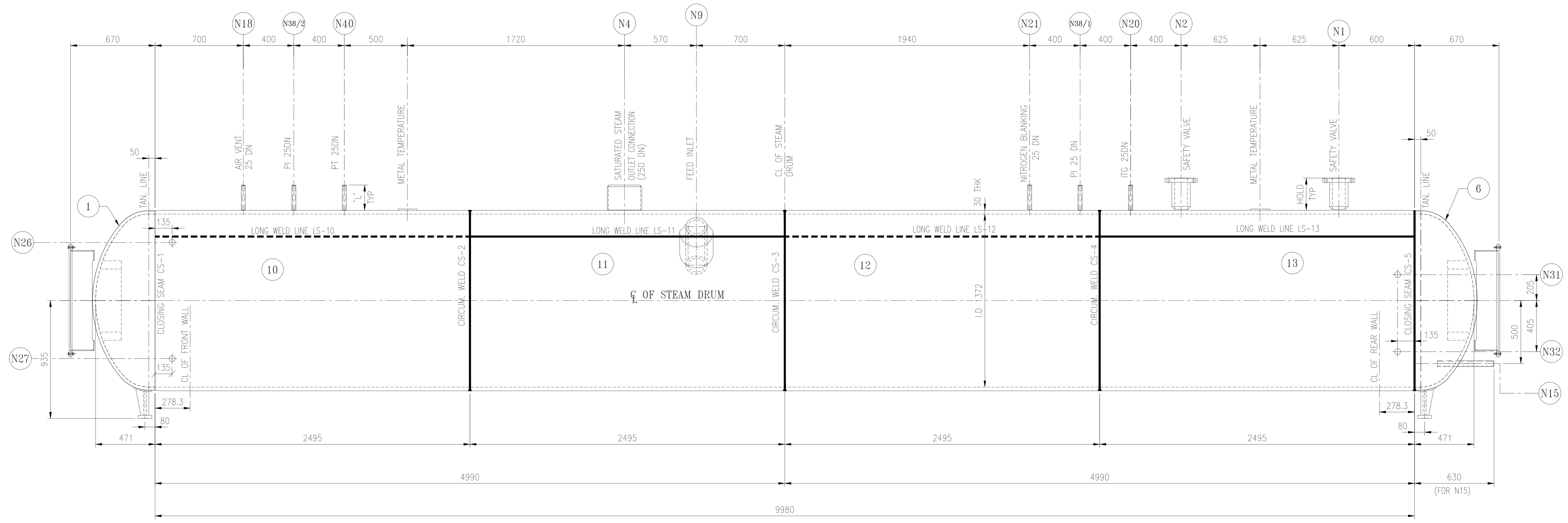
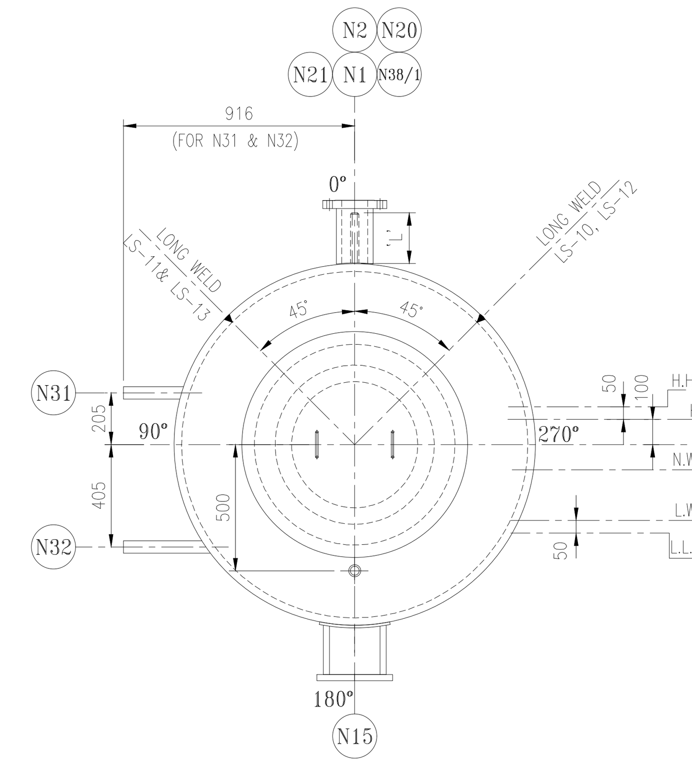
DOCUMENT ID	JB1178-50010100-007-0001
PROJECT ID	JB1178



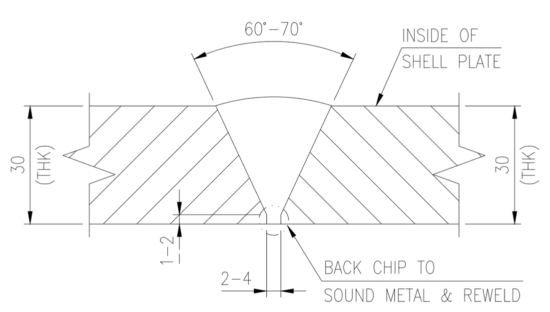
VIEW FROM BOILER FRONT SIDE



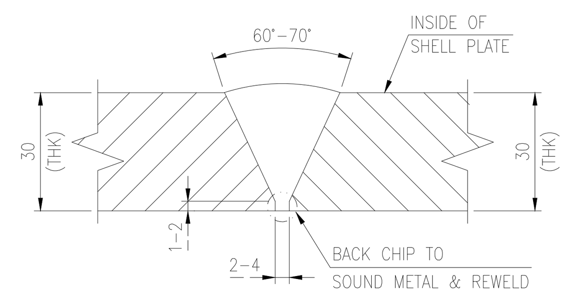
ARRANGEMENT OF STEAM DRUM
ELEVATION FROM R.H.S. OF BOILER



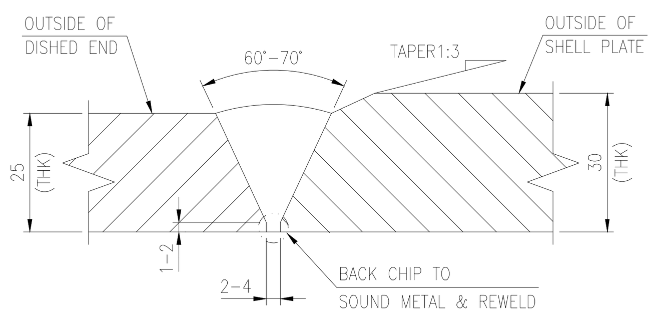
VIEW FROM BOILER REAR SIDE



WELD PREPARATION OF LONGITUDINAL SEAMS OF DRUM
FOR SEAM MK.Nos. LS-10 TO LS-13

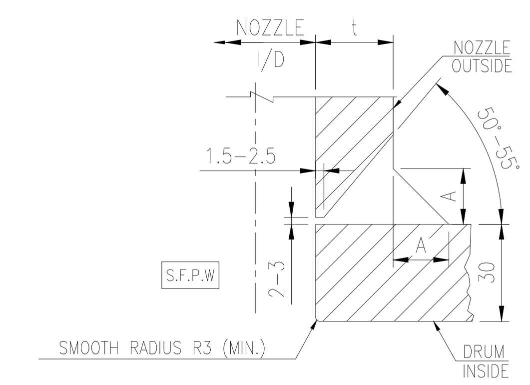


WELD PREPARATION OF CIRCUMFERENTIAL SEAMS OF DRUM
FOR SEAM MK.Nos. CS-2, CS-3 & CS-4



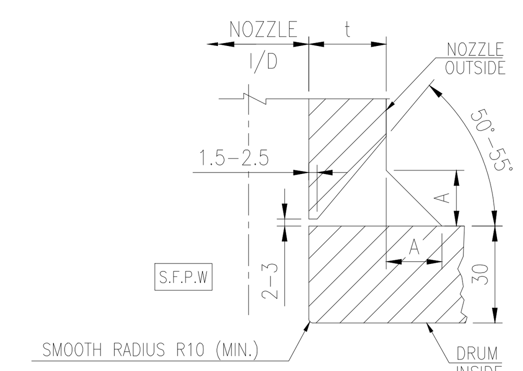
WELD PREPARATION OF CLOSING SEAM OF DRUM
FOR SEAM MK.Nos. CS-1 & CS-5

IBR CHAPTER - XII
IBR FIG : XII/85E B2



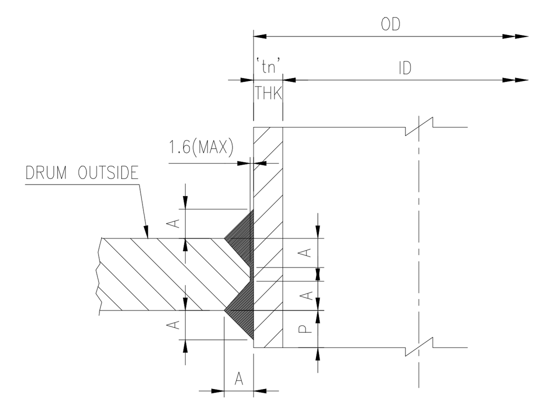
WELD DETAIL OF NOZZLE TO DRUM
(NOZZLE MARK. NO.- N4, N9, N18, N20, N21 N26, N27, N31, N32, N38/1, N38/2 & N40)

IBR CHAPTER - XII
IBR FIG : XII/85E B2

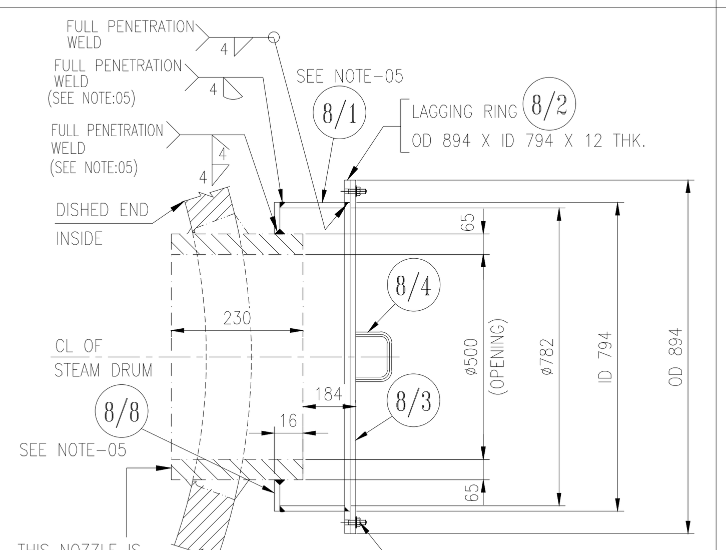


WELD DETAIL OF NOZZLE TO DRUM HOLD
(NOZZLE MARK. NO.- N1 & N2)

IBR FIG : 365/13
CHAPTER : VIII

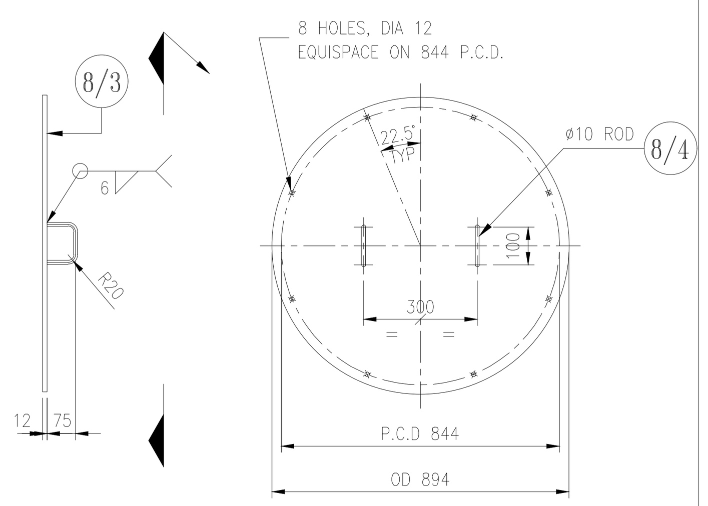


WELDING DETAIL OF NOZZLE
(NOZZLE MARK. NO- N15
WELDING DETAIL

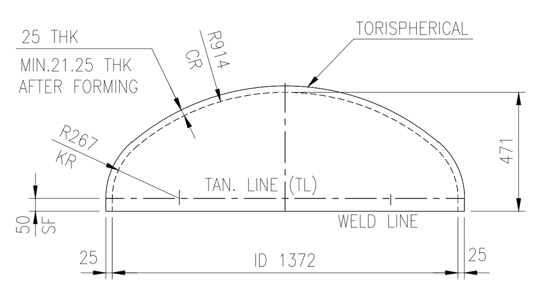


THIS NOZZLE IS RELEASED IN DRG.NO. JB1178-50010100-006-0001

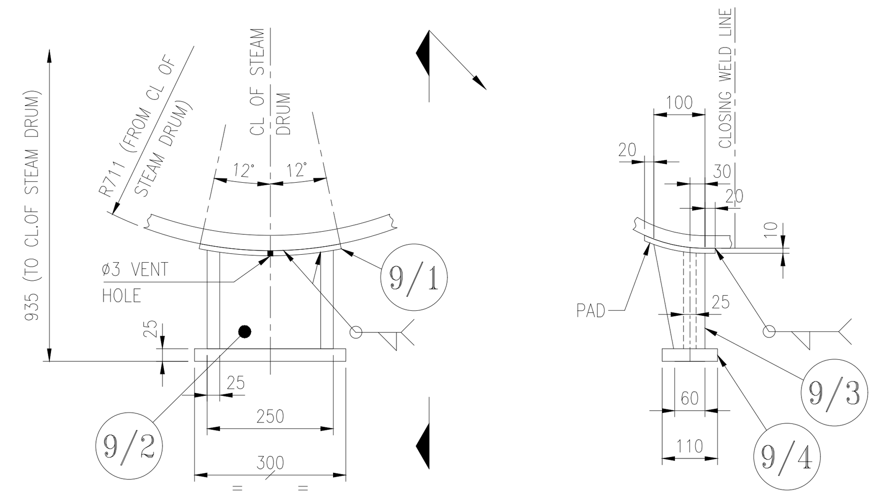
DETAIL OF MANHOLE OPENING



DETAIL OF MANHOLE COVER



STEAM DRUM DISHED END DETAIL



[ALL WELDS ARE 10mm FILLET WELD]

DETAIL OF GUIDE BRACKET IN STEAM DRUM

DESIGN DATA

CODE OF CONSTRUCTION	I.B.R. (1950) AND IT'S AMMENDMENTS UPTO AUGUST 2020 (SEE NOTE:01)
MAXIMUM WORKING PRESSURE	26.5 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	228°C
WORKING METAL TEMPERATURE	256°C [INSIDE GAS PATH]
HYDRO TEST PRESSURE	39.75 Kg/Sq. cm (g) AS PER IBR AT SHOP & SITE
RADIOGRAPHY OF LONG & CIRCUM SEAM	100%
STRESS RELIEVING	YES REFER HEAT TREATMENT CYCLE TABLE
EVAPORATION OF BOILER (M.C.R.)	72.30 TPH
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING ECONOMISER & SUPER HEATER)	1005 Sq.m (APPROX.)
HEATING SURFACE AREA OF ECONOMISER (NON STEAMING)	NOT APPLICABLE
HEATING SURFACE AREA OF DRUMS	22 Sq.m (APPROX.)
FLANGE PARTICULAR UNLESS STATED OTHERWISE	REFER DRG. NO.- JB1178-50010100-009-0001.
STEAM PRESSURE AT SUPER HEATER OUTLET	19.00 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	310±5°C
WEIGHT OF DRUMS	12880.74 Kgs (EMPTY)
INSPECTION	CIB (HARYANA) SEE NOTE: 02

TABLE - 1

S.Nos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (D) mm	Thickness of Nozzle	Nozzle Thickness (tn) mm	Connecting Pipe Schedule/ Thickness (mm)	Projection from OD of Drum (L) mm	Nozzle CL to Drum Vertical Axis (±X) mm	Nozzle CL to Drum Horizontal Axis (±Y) mm	Calculated Nozzle Length (R) mm	Finished Length of the Nozzle (M) mm	M/C Bore for end Preparation (E) mm	Weld Dim. (A) mm	Prjn - In from cl of nozzle (p) mm
1	N1	SAFETY VALVE NOZZLE-1	STEAM DRUM SHELL	1						HOLD						0
2	N2	SAFETY VALVE NOZZLE-2	STEAM DRUM SHELL	1						HOLD						0
3	N4	SATURATED STEAM STUB	STEAM DRUM SHELL	1	273.1	SCH.80	15.09	7.1 THK.	200	–	–	240	207	258.9	15	0
4	N9	THERMAL SLEEVE FOR FEED INLET NOZZLE	STEAM DRUM SHELL	1	219.1	SCH 120	18.26	–	–	–	–	REF.DRG: -009-0001				125
5	N15	CBD NOZZLE	STEAM DRUM DISH	1	48.3	SCH.XXS	10.15	SCH.40	–	–	500	480	450	40.94	10	142
6	N18	AIR VENT NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	–	–	230	200	26.64	10	0
7	N20	INSTRUMENT TEST GAUGE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	–	–	230	200	26.64	10	0
8	N21	NITROGEN BLANKETING NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	–	–	230	200	26.64	10	0
9	N26	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	–	–	460	420	389	40.94	11	0
10	N27	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	–	–	460	420	389	40.94	11	0
11	N31	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	–	–	205	270	238	40.94	11	0
12	N32	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	–	–	405	375	343	40.94	11	0
13	N38/1&2	PRESSURE GAUGE NOZZLE (LOCAL)	STEAM DRUM SHELL	2	33.4	SCH.XXS	9.09	SCH.40	200	–	–	230	200	26.64	10	0
14	N40	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	–	–	230	200	26.64	10	0

NOTES :-

01. THE ZERO DATE OF THIS CONTRACT IS 01.08.2020 HENCE IBR AMENDMENTS UPTO AUGUST 2020 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
02. INSPECTION BY CIB HARYANA BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL BE DONE AT ISGEC WORKS, YAMUNA NAGAR (HARYANA)
03. ALL DIMENSIONS ARE IN 'mm' UNLESS OTHERWISE SPECIFIED.
04. FOR GENERAL NOTES & MATERIAL SPECIFICATION REF. DOCUMENT ID : JB1178-50010100-007-0002.
05. P. NO. 8/8 RING PLATE FIRST TO BE WELDED WITH MANHOLE NOZZLE AFTER P. NO. 8/1 (CLEAT FOR RING) WELDED WITH P. NO. 8/8 (RING PLATE).

MATERIAL SPECIFICATIONS :

DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.cm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	28.03	
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	19.53	

LIST OF REFERENCE DRAWINGS

DRAWING DOCUMENT ID	DESCRIPTION
JB1178-50010100-006-0001	ARRANGEMENT & DETAIL OF CIRCULAR MANHOLE ON STEAM DRUM
JB1178-50010100-007-0002	DETAIL AND MATERIAL LIST OF STEAM DRUM
JB1178-50010100-008-0001	LAYOUT OF TUBE HOLES ON STEAM DRUM
JB1178-50010100-009-0001	ARRANGEMENT & DETAIL OF STEAM DRUM NOZZLES

REVN.	ZONE	DESCRIPTION	REVD. BY	CHKD. BY	APPD. BY	DATE	REVN.	ZONE	DESCRIPTION	REVD. BY	CHKD. BY	APPD. BY	DATE

Issued to _______ Issued by _______ Issued date _______

☐ For pre-planning ☐ For manufacture in shop _______
☐ For provisional manufacture in shop _______ ☐ For sub contracting / site work
This drawing supersedes drawing no. ___________ Rev. ___

W.O. No. PD - 0353 / I.W.T. No. : 6677

CONSULTANT	--
CLIENT	M/s. L&T MHPS BOILERS PVT LTD., KHURJA , UTTAR PRADESH.

DO NOT SCALE
IF IN DOUBT PLEASE ASK

FOR APPROVAL	NO	FOR INFORMATION	NO	FOR PRODUCTION	YES	FOR ERECTION	YES

RESPONSIBLE DEPT.	DRAWN BY	CHECKED BY	APPROVED BY	DATE
MECHANICAL	V.KAVASKAR	R.G.AMIR/R.KUMARAN	R.ARUNKUMAR	17.09.2020
	TOTAL WEIGHT IN kg. REF.007-0002	PROJECT ID JB1178	ELEMENT ID 50010100	I.W.T. NO. 6677
		YEAR : 2020	SHEET SIZE : A1	SCALE :1 : 25

ISGEC
BOILERS
(formerly Isgec John Thompson)
Isgec Heavy Engineering Ltd.
Noida 201 301 (U.P.) India

SERVICE	BI DRUM, WATER TUBE, BOTTOM SUPPORTED OIL FIRED AUXILIARY BOILER
TITLE	ARRANGEMENT AND DETAIL OF STEAM DRUM
DRAWING DOCUMENT ID	JB1178-50010100-007-0001 REVISION 00